Thermal equilibrium does not depends on thermal conductivity of the material of bodies X and Y.

13 *Ans:* ***B***

Work done on gas is positive during compression, hence work done by gas is negative during compression (Y to Z).

For ideal gas, U = 3/2 pV. Hence the internal energy of the ideal gas decreases with constant pressure but decreasing volume.
With work done on gas being positive, the heat is transferred out of the gas, i.e. heat is removed.

14 *Ans:* ***C***